

Features

- It comprises various objects which can be used readily — text objects, 3D boxes, own customised shapes via Shape Painter, the particle engine, dynamic lights and shadows, custom collision masks, etc.
- Adds behaviours to the objects through the physics engine, pathfinding, top-down movement, platformer engine, draggable objects and the automation of tasks.
- Offers advanced design features and interfaces through the scene editor, multiple layers, the debugger and performance profilers.
- Other features include HTML 5 support, sound and music effects, and integration with the joystick and keyboard.

Latest version: 4.0.94

Official website: <http://compilgames.net/>



Figure 1: GDevelop user interface

Godot Engine

The Godot Engine is a highly powerful cross-platform game development engine that supports 2D and 3D game development from a unified interface. The platform supports Windows, OS X, Linux and BSD for developing games for the PC, console and even mobile-cum-Web platforms. It is integrated with a wide variety of tools, providing developers with tons of options, and avoiding the need for even a single third party tool. The engine is built on the concept of a tree of nested scenes.

The games created with Godot are either written in C++ or a customised scripting language called GDScript, which is a high level, dynamically typed language with many similarities to Python. GDScript is greatly customised and optimised for the Godot engine. Godot has a power text editor, which provides developers various features like auto indentation, highlighting syntax and even code completion. It also has a debugger to provide breakpoints and program stepping.

Godot makes use of the OpenGL ES 2.0 graphics engine, which has many features like transparency, normal mapping, dynamic shadows using shadow maps, and various post-processing effects like FXAA, bloom, DOF, HDR, gamma correction and fog.

Features

- *Nice and clean interface:* Has a visual editor, a dynamic scene system, a user friendly content creation interface, a visual shader editing tool and live editing on mobile devices.
- Efficient in 2D game design because of a dedicated 2D engine, custom 2D physics engine, and a flexible kinematic controller.
- High-end 3D game development by importing animation models from 3DS Max, Maya and Blender; has skeleton deforms and blend shapes, lighting and shadow mapping, HDR rendering, anti-aliasing, etc.
- Flexible animation engine for games, enabled by the visual animation editor, frame-based or cut-out animation, custom transition curves and tweens, and animation tree support.
- Other features include a Python-like scripting language, a powerful debugger and an easy C++ API.

Latest version: 2.1.3

Official website: <https://godotengine.org>

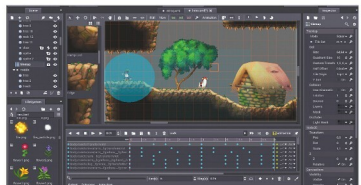


Figure 2: Godot Engine interface

Cocos2d-x

Cocos2d-x is an open source game development platform available under the MIT License. It allows developers to build games, apps and various interactive programs. It enables developers to make use of C++, Lua and JavaScript for cross-platform deployment on iOS, Android, Windows Phone, OS X, Windows and Linux devices.

The cocos2d-x renderer engine is highly optimised for 2D graphics with OpenGL support. It is packed with tons of features like skeletal animation, sprite sheet animation, coordinate systems, visual effects, textures, tile maps, multi-resolution devices, etc.

It is maintained by developers at Chukong Technologies, which is also developing Cocostudio, a WYSIWYG editor.

Features

- *Animation:* It provides numerous animation options that work on sprites using a set of actions and timers. It supports animation of particle effects, image filtering effects through shaders, etc.
- *Easy GUI:* It includes an easy GUI interface with text